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https://github.com/antoniomachuca/conway-99-graph

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To:
The Editors-in-Chief and Editorial Board
The Electronic Journal of Combinatorics

Subject: Submission of the manuscript "Symmetry-Restricted SAT Encodings, Lean 4 Formalization, and Verification Limits for Conway's 99-Graph"

Dear Editors,

Please find enclosed the manuscript entitled **"Symmetry-Restricted SAT Encodings, Lean 4 Formalization, and Verification Limits for Conway's 99-Graph"**, which I submit for consideration for publication in *The Electronic Journal of Combinatorics*.

The manuscript concerns the existence of a strongly regular graph with parameters $\mathrm{srg}(99, 14, 1, 2)$, an open problem on J. H. Conway's list of five \$1,000 problems. In the cited literature, two symmetric cases remain open: an involution with exactly one fixed vertex and a fixed-point-free automorphism of order 3. The manuscript does not decide existence or rigidity. It records which combinatorial and computer-assisted checks were carried out and what each check establishes:

1. **Mechanized structural lemmas.** Lean 4 proofs of K_4-freeness, the $7K_2$ decomposition of every neighborhood, diameter at most 2, and a computable checker proved equivalent to the adjacency predicate; the kernel reports only the axioms `[propext, Quot.sound]`. These facts are standard; the contribution is their mechanization. Arithmetic contradictions from character-theoretic arguments are also kernel-checked; the graph-level wrappers assume their premises and are not reported as proved.
2. **A mechanized order-7 reduction.** From a Conway 99-graph and a nontrivial automorphism s with $s^7 = \mathrm{id}$, a Lean 4 proof constructs a canonical 15×15 orbit quotient matrix. It derives the fixed vertex, the orbits, and the quotient equations; it does not assume them. The kernel reports only `[propext, Quot.sound]`. The orbit structure is known (Behbahani–Lam, Cesarz–Woldar); the contribution is the mechanization. Lean does not prove that no such matrix exists.
3. **The involution case with one fixed vertex.** Every binary 7×42 matrix with $CC^T = 10I_7 + 2J_7$ has column weight 2. The support labels of the three branch representatives are corrected; their classes have sizes 1, 20, and 20 under the stabilizer of a distinguished orbit.
4. **A checker-verified order-7 CNF refutation.** For one canonical order-7 CNF (177,097 variables, 422,508 clauses), CaDiCaL 1.9.5 returns `s UNSATISFIABLE`, and `drat-trim` reports `s VERIFIED` on the 26,746,488,053-byte DRAT proof. Behbahani and Lam (2011) already excluded order 7. The graph-to-CNF translation is not kernel-verified, so the manuscript claims no new proof of that exclusion. It does not decide the CP-SAT model of Thakkar and Severini (2026).
5. **Exterior-row CNF, separate from the Lean reduction.** The exterior-row CNF has 26,166 variables and 311,703 clauses. It is not the canonical order-7 CNF, and it is not in the Zenodo record. CaDiCaL 1.9.5 returned `s UNSATISFIABLE`, and `drat-trim` reported `s VERIFIED`. That refutation is PROVED for this CNF only. The Lean derivation of the exterior equations is a different PROVED component. The implication from `Z7OrbitMatrix` to the CNF is PENDING. The DRAT proof is not checked inside the Lean kernel, and Lean does not prove that no orbit matrix exists.
6. **An inconclusive search and positive controls.** A 12.5-day CaDiCaL campaign on the open involution and order-3 encodings (more than 2.4×10^9 conflicts, recorded aggregate) found no model and no refutation; it started before a compiler repair. Positive controls apply the production compilers for these two actions to Paley(9) and to the Berlekamp–van Lint–Seidel graph $\mathrm{srg}(243, 22, 1, 2)$. They do not prove compiler correctness.

Every claim carries one of four labels (PROVED, COMPILED, EXPLORED, PENDING) defined in the manuscript. The Lean 4 library, the SAT compilers, and the regression tests are available on GitHub (`https://github.com/antoniomachuca/conway-99-graph`). The canonical order-7 CNF and its DRAT proof are in Zenodo record 10.5281/zenodo.22995101. The solver log, the checker log, and the SHA-256 digests for that CNF are in the same record. The exterior-row CNF is not in that record.

Declarations: The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests.

Suggested Reviewers: I suggest the following three referees:

1. **Prof. Dean Crnković** (University of Rijeka, `deanc@math.uniri.hr`) — Strongly regular graphs and automorphism groups; co-author of the exclusion of automorphism groups of order 6 and 9 for $\mathrm{srg}(99, 14, 1, 2)$.
2. **Prof. Marijn J. H. Heule** (Carnegie Mellon University, `marijn@cmu.edu`) — Computer-assisted proofs in

combinatorics, SAT encodings, clausal proof checking.

3. **Prof. Simone Severini** (University College London, `s.severini@ucl.ac.uk`) — Algebraic graph theory; constraint formulations for Conway's 99-graph.

Thank you for your time and consideration of this work.

Sincerely,

Antonio Machuca